



MODULATION

What are modulated wave? why are they required?

Modulated wave: In the transmission of speech and music by wireless, we have the problem of converting sounds of a fairly low frequency (audio frequency) into some electrical waves and of transmitting this wave over large distances and of reconverting it into sound.

It is possible to convert the sound into an a.c. of the same frequency and to use this to produce electromagnetic waves for no radiation of appreciable energy occurs from an antenna in which currents of audio frequency are flowing.

It is necessary therefore to use high frequency currents which produce a high frequency wave and to superimpose upon, the effect of the audio frequency not

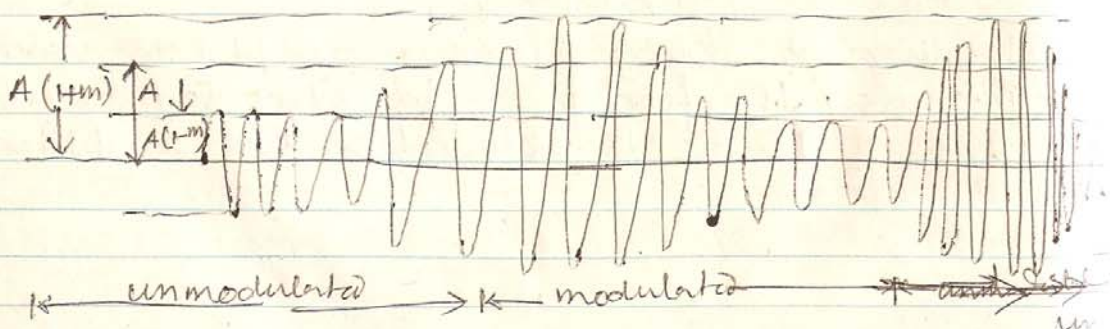


Fig: (1)

The method actually consists of transmitting high frequency wave ($f = \omega_c$) and varying its amplitude sinusoidal with frequency ω_m of



an audible note. Such a wave is illustrated in fig:1 and is called an "Amplitude Modulated wave" the high frequency being called the "Carrier wave". The portions on the right and left of fig:1 represent the unmodulated carrier wave of amplitude A . When the modulation is applied the amplitude is varied from the maximum value $\{A(1+m)\}$ to the ~~minimum~~ minimum $\{A(1-m)\}$. The quantity 'm' represents the magnitude of the modulation effect and when expressed as a percentage, it is known as the "Percentage Modulation", m always lies between 0 and 1.

This type of modulation is known as "Amplitude modulation" because the amplitude of the high frequency oscillation is varied at the audio frequency. In the other type of modulation, the amplitude of the high frequency oscillation remains constant but its frequency is varied periodically at audio frequency. The latter is called "Frequency Modulation".

Modulation may be defined as the process by which some characteristics of a relatively high frequency wave (called the Carrier) is varied in accordance with the instantaneous value of a low frequency wave (called the modulating wave). Hence, the information i.e. speech or music is the modulating signal.

Theory Of Amplitude Modulation:

Amplitude modulation is defined as one in which the amplitude of the carrier wave (h.f wave) is ~~not~~ proportional to the instantaneous amplitude of the modulating voltage (having audio frequencies)

Let,
$$e_c = E_c \sin \omega_c t \quad \text{--- (1)}$$

$$e_m = E_m \sin \omega_m t \quad \text{--- (2)}$$

Modulation



where E_c = Carrier voltage.
& E_m = modulating voltage.
 ω_m = angular frequency of the modulating signal.

In both eqⁿ ① and ② the phase-angle has been ignored.

From the definition of AM (amplitude modulation) it follows that the amplitude E_c of the unmodulated (h.f.) carrier will have to be made proportional to the instantaneous modulating voltage E_m when the carrier is amplitude modulated.

$$\text{further, } m = \frac{E_m}{E_c} \quad \text{--- (3)}$$

where, m = modulation index | depth of modulation

Then if A = resulting amplitude of the A.M wave voltage.

We have,

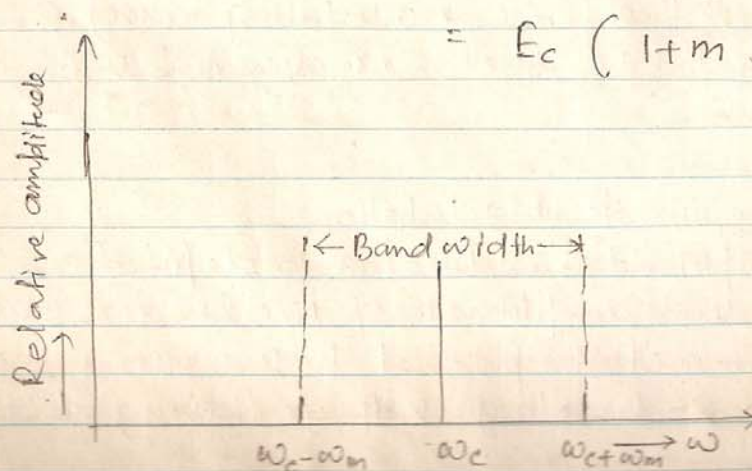
$$A = E_c + E_m$$

using eqⁿ ② & ③ we get

$$A = E_c + E_m \sin \omega_m t$$

$$= E_c + m E_c \sin \omega_m t$$

$$= E_c (1 + m \sin \omega_m t) \quad \text{--- (4)}$$





The instantaneous voltage of the resulting AM wave according to eqⁿ (4) is given by

$$e = A \sin \theta = A \sin \omega_c t$$

using eqⁿ (4), therefore

$$e = E_c (1 + m \sin \omega_m t) \cdot \sin \omega_c t \quad \text{--- (5)}$$

$$\text{or } e = E_c \sin \omega_c t + E_c \cdot m \cdot \sin \omega_m t \cdot \sin \omega_c t$$

using the identity

$$\sin x \cdot \sin y = \frac{1}{2} \{ \cos(x-y) - \cos(x+y) \}$$

$$\therefore e = \left[E_c \sin \omega_c t + \frac{E_c \cdot m}{2} \{ \cos(\omega_c - \omega_m) - \cos(\omega_c + \omega_m) \} \right] \quad \text{--- (6)}$$

(In practice ω_m is small compared to ω_c)

Equation (6) is the eqⁿ of an AM wave which contains three terms. The first term represents the unmodulated carrier. The two additional terms produced are the two side bands.

Frequency Spectrum: Eqⁿ (6) shows that an A.M. carrier with sinusoidal modulation consists of three frequencies ω_c , $(\omega_c + \omega_m)$ & $(\omega_c - \omega_m)$. The frequency $(\omega_c + \omega_m)$ is called the upper side frequency and $(\omega_c - \omega_m)$ is called the lower side frequency.

Fig: 2 gives the relative amplitudes frequency components; such a plot is known as the frequency spectrum. $(BW)_{AM} = 2 f_m$ highest

m_a is defined as the ratio of the maximum deviation of amplitude from the modulated value (E) to the unmodulated carrier amplitude. It is often expressed as a percentage.

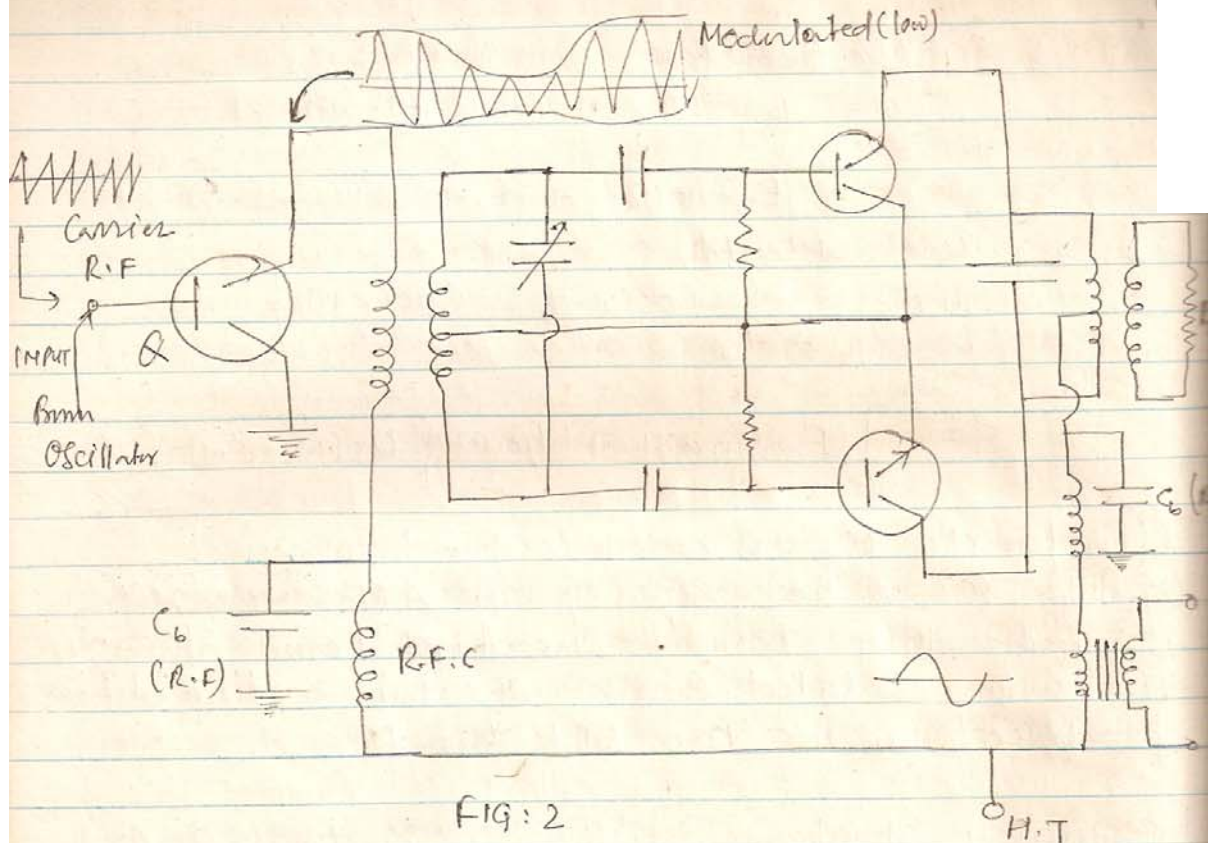
$$\text{Percentage modulation} = m_a \times 100$$

m always lies between 0 & 1.

Modulation



Modulated Transistor Amplitude Circuit
At the Lower power levels is shown below



Operation: High frequency Oscillations are generated in a separate Oscillator and fed into the base of the modulating transistor (Q). Its collector current therefore Oscillates at the same h.f. Thus the carrier is produced.

Now, through a transformer transformer (T_n), the audio signal is applied to the collector of the same transistor resulting in the generation of the modulated wave over the impedance in its collector circuit. This is inductively coupled to the bases of the push pull stage for power amplification. The final output is

Modulation



collected inductively across the load.

The combination of radio frequency coil (RFC) and the by-pass capacitor (C_b) is to confine the h.f. currents to the transformer & and isolate the transformer secondary of Tm.

② FREQUENCY MODULATION:

In F.M the instantaneous frequency of the carrier is varied by the modulating signal. In this process the carrier wave is modulated such that the angular frequency (ω) is varied in time in accordance with the modulating signal amplitude (in voltage) is

$$\omega = \omega_0 + K_f V_m \cos \omega_m t \quad \text{--- (1)}$$

Where, $\omega_0 = 2\pi f_0$ is called "the central frequency".

ω_m = the modulating angular frequency.

and $K_f V_m$ = is constant of proportionality

Degree of frequency variation is proportional to the modulating amplitude.

Again,

the instantaneous voltage amplitude may be written as,

$$V = V_0 \cos \left(\int \omega dt + \phi_0 \right) \quad \text{--- (2)}$$

from (1) and (2) :

$$V = V_0 \cos \left\{ \int (\omega_0 + K_f \omega_m \cos \omega_m t) dt \right\}$$

(having dropped ϕ_0)

where the time t is measured from the reference point at which $V = 0$

$$\therefore V = V_0 \cos \left\{ \omega_0 t + \frac{K_f V_m \sin \omega_m t}{\omega_m} \right\} \quad \text{--- (3)}$$

The maximum value of $\left(\frac{K_f V_m}{2\pi} \right)$ is called the frequency deviation :

$$(\Delta f)_{\max} = \frac{K_f V_m}{2\pi} \quad \text{--- (A) and}$$

is denoted by Δf . This is the $^{2\pi}$ maximum deviation of frequency in Hz. from the central frequency f_0 .



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Modulation



The ratio of $(\Delta f)_{\max}$ to the maximum modulating frequency $(f_m)_{\max}$ is called the "Deviation Ratio" 'mf' of the system.

$$\text{i.e. } mf = \frac{(\Delta f)_{\max}}{(f_m)_{\max}} = \frac{K_f \cdot V_m}{\omega_m} \quad \text{--- (4)}$$

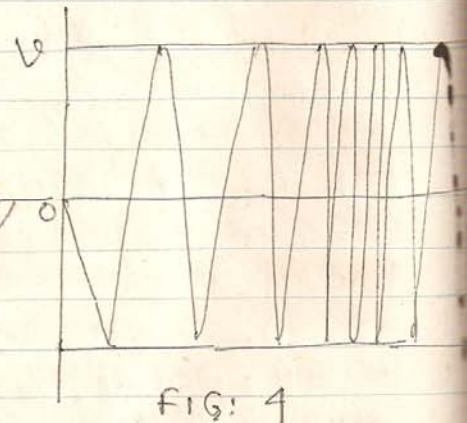
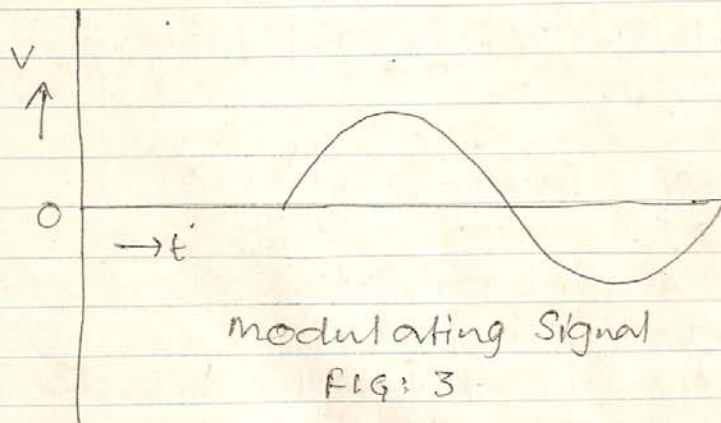
Finally, using eqⁿ (3) & (4) the frequency modulated wave may be written as below.

$$V = V_0 \cos \{ \omega_c t + mf \sin \omega_m t \} \quad \text{--- (5)}$$

From eqⁿ (4) we see that:

The maximum frequency deviation $(\Delta f)_{\max}$.

The frequency deviates from the control frequency is directly proportional to the amplitude of the modulating signal. But the amount of deviation is independent of the modulating frequency and the F.M modulated wave is of constant amplitude as shown in fig below.



F. M Carrier wave.

Frequency Spectrum for F.M:

Equation (5) is of the form $\cos(a+b)$ and can be expanded to,

$$\begin{aligned} & \cos \{ \omega_c t + mf \sin \omega_m t \} \\ &= \{ \cos \omega_c t \cdot \cos (mf \sin \omega_m t) - \sin \omega_c t \cdot \sin (mf \sin \omega_m t) \} \end{aligned}$$



The 2nd. factor in each term may be expanded into an infinite series as given below:

$$\cos(m_f \sin \omega_m t) = J_0(m_f) + 2J_2(m_f) \cos 2\omega_m t + 2J_4(m_f) \cos 4\omega_m t + \dots$$

and

$$\sin(m_f \sin \omega_m t) = 2J_1(m_f) \sin \omega_m t + 2J_3(m_f) \sin 3\omega_m t + 2J_5(m_f) \sin 5\omega_m t + \dots$$

The J_n coefficients are Bessel's function of the 1st kind and order n , defined by an infinite series and tabulated in standard mathematical tables. Using the identities,

$$\cos a \cdot \cos b = \frac{1}{2} [\cos(a+b) + \cos(a-b)]$$

and

$$\sin a \cdot \sin b = \frac{1}{2} [\cos(a-b) - \cos(a+b)]$$

We can write an expression for the components of a frequency modulated wave as,

$$V = V_0 [J_0(m_f) \cos \omega_0 t - J_1(m_f) \{ \cos(\omega_0 - \omega_m)t - \cos(\omega_0 + \omega_m)t \}$$

$$+ J_2(m_f) \{ \cos(\omega_0 - 2\omega_m)t + \cos(\omega_0 + 2\omega_m)t \} \\ - J_3(m_f) \{ \cos(\omega_0 - 3\omega_m)t - \cos(\omega_0 + 3\omega_m)t \} \\ + J_4(m_f) \{ \cos(\omega_0 - 4\omega_m)t + \cos(\omega_0 + 4\omega_m)t \} + \dots] \quad (7)$$



Generally we do not need to work with infinitely wide frequency band. For $n > m_f > 5$ the coefficient $J_n(m_f)$ decrease rapidly. The needed F.M band width can be approximated as.

$$B_w \approx 2m_f \cdot f_{m_{\max}} = 2\Delta f_{\max} \quad (8) \quad | \quad f_{m_{\max}} \quad (9)$$



(1) Diode FM Generation:

An FM Signal may be generated by the variation of the capacitance of a tuned circuit oscillator by the modulating signal. This can be achieved by connecting a reverse biased diode across the oscillator tuned circuit as shown in fig: 5.

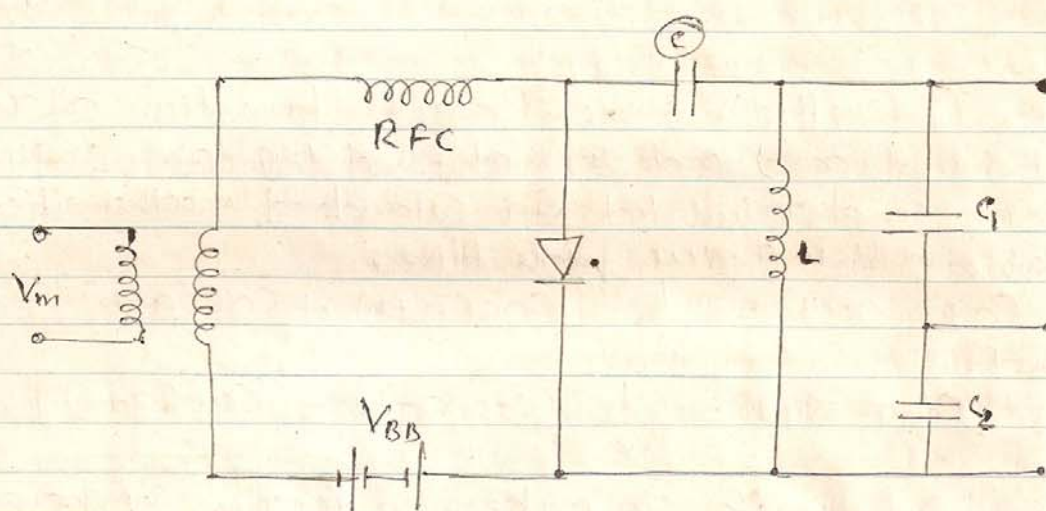


FIG: 5 Diode FM Modulator.

The capacitance C of the junction will vary if the modulation is added to the bias voltage as:

$$V_B = -V_{BB} + V_m \cos \omega_m t \quad \text{--- (9)}$$

With the capacitance of the diode proportional to the square root of the applied voltage and with V_m kept small and approximate linear change in frequency can be obtained from sine

$$\omega = \frac{1}{L(C + K V_B^{1/2})} \quad \text{--- (10)}$$

(2) Detection of FM Signals: Detection of



FM Signals is preceded by an (Amplitude Limiter) which is described below:

Detection: Detection is also called demodulation is the reverse process of modulation i.e. in the process of detection the modulating signal is recovered from the modulated carrier.

The information or the intelligence in the form of modulating signal is superimposed on the carrier at the transmitting station for the case of transmission. At the receiver the intelligence is recovered by separating the modulating signal from the modulated carrier and discarding the carrier.]

II

(A) Detector:

(A) Emitter-Coupled Limiter:-

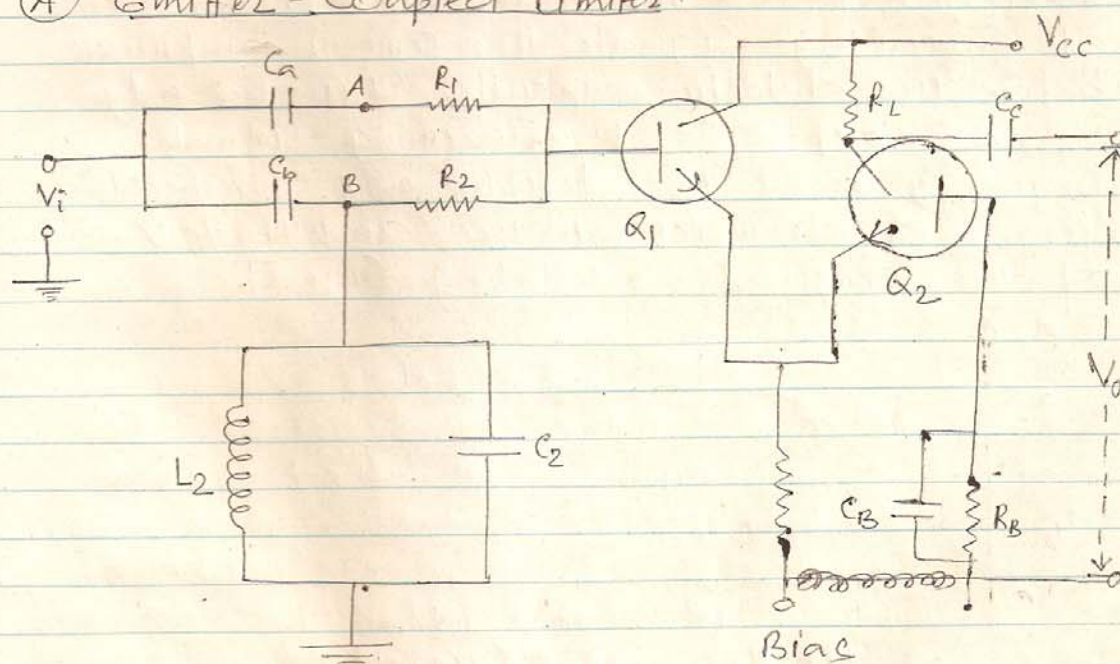


FIG: (6) Emitter-Coupled Limiter. (detector)

(3) FM Detector: The received FM signal is pass through an emitter coupled limiter shown in FIG: 6. The output is fed to a



Foster-Seeley discriminator Shown in FIG: 7 which ~~(Fig 7)~~ converts the frequency variation into amplitude variation i.e. an FM detector first converts the incoming signal to a constant amplitude signal into a signal whose amplitude varies with frequency i.e. it converts FM into AM signal. finally an A.M detector de-modulates this AM signal and recovers the original audio-signal.]

①. The Amplitude Limiter:

Noise impulse and amplitude modulated signal may arrive with F.M signal as interference. Since FM signals have constant amplitude, the interference may be removed by reducing all incoming signals to a common amplitude before the detection circuits. Then the only varying property of the incoming signal is frequency and the detector will convert the frequency deviations to the amplitude variation of the original modulating signal.

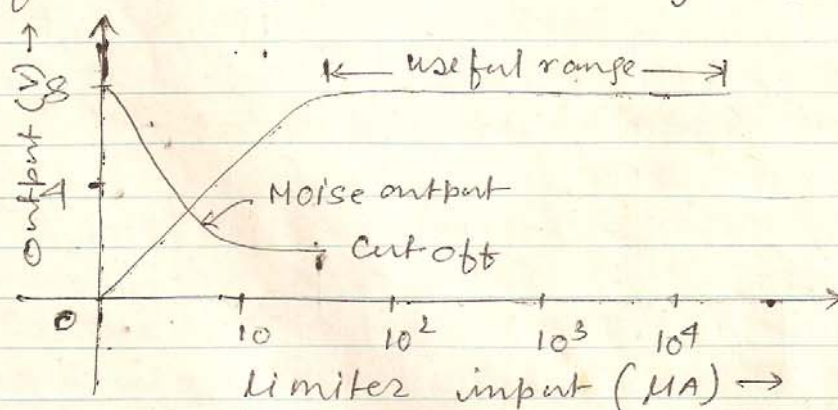


FIG: 1

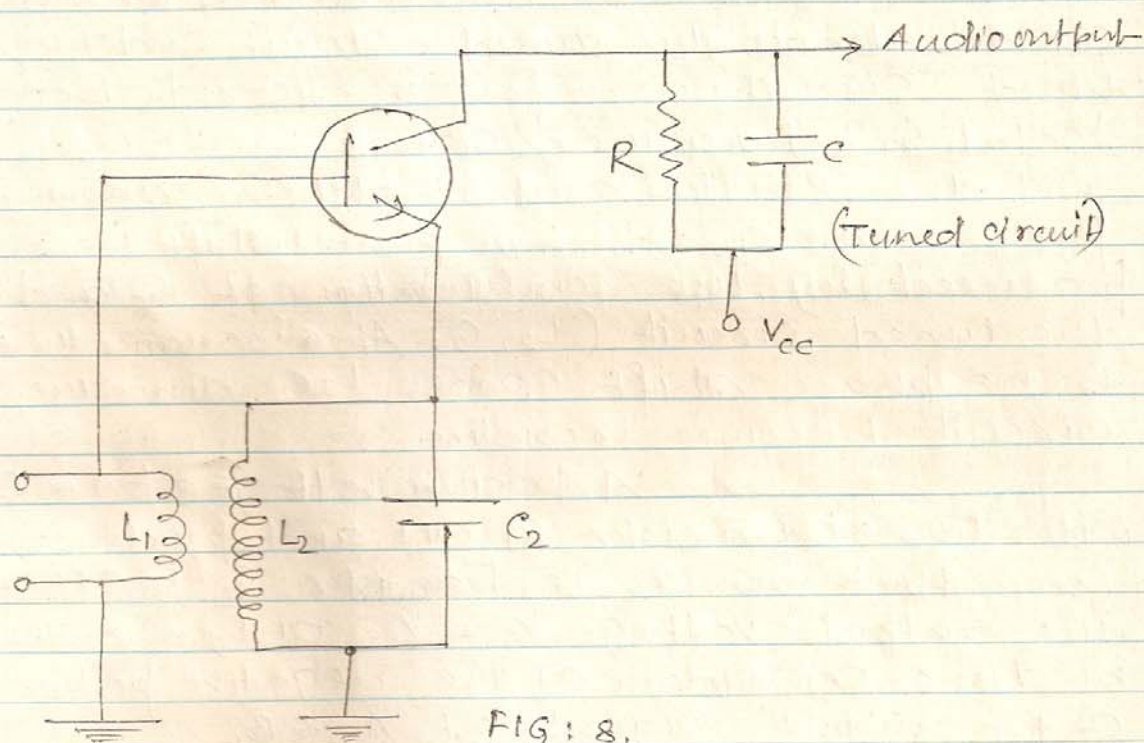
The function of limiting signals to a common amplitude can be performed by a transistor operated at low collector voltage. All incoming signals then drive the transistor



between cut off and a low value of saturation voltage and a constant output amplitude is maintain as shown in fig: 4.]

D. Alternative Detector for FM Signal:

The Quadrature (there is a phase diff. of 90°) Detector for FM



A Simplified quadrature detector for FM signal is shown in fig: 8.

The transistor operates into cut off and saturation and serves as its own amplitude limiter.

With the transformer secondary tuned to f_0 (centre frequency), the secondary voltage across $L_2 C_2$ is in quadrature w.r. to the primary voltage and the base voltage.

FIG: 8 shows the principle of the quadrature detector. The length of time



that the base is positive to the emitter (when the Transistor is conducting) varies with the incoming frequency. The saturation current but vary in length and the average output voltage across 'c' changes in accordance with the modulation.

Because of the heavy current demand of the base in saturation, the base circuit loads the tuned circuit and the tuned circuit wave form is distorted. Better isolation between these signals is obtained with the emitter coupled detector shown in fig: 6. The quadrature phase shift is produced by the combination of C_6 and the tuned circuit (L_2, C_2). At resonance there is no phase shift across L_2C_2 since the circuit becomes resistive.

A positive voltage at A or B will turn transistor Q_1 ON and produce zero output from Q_2 i.e. transistor Q_2 is OFF and the output voltage $V_o = V_{cc}$ only when there is time-coincidence of the negative portions of the input signals at A & B.

Resistor R_1 & R_2 isolate the voltage at A from the tuned circuit at B.

The tuned circuit is designed for high Q with a relatively large value of capacitance (500pF at $f_o = 10^6$ Hz). Large Q gives increased output voltage per cycle of frequency variation.

At resonance the secondary impedance is pure resistive.

We may consider C_2 to be made up of two equal capacitors in series. Hence the phase angle of i_{ep} differs from that of

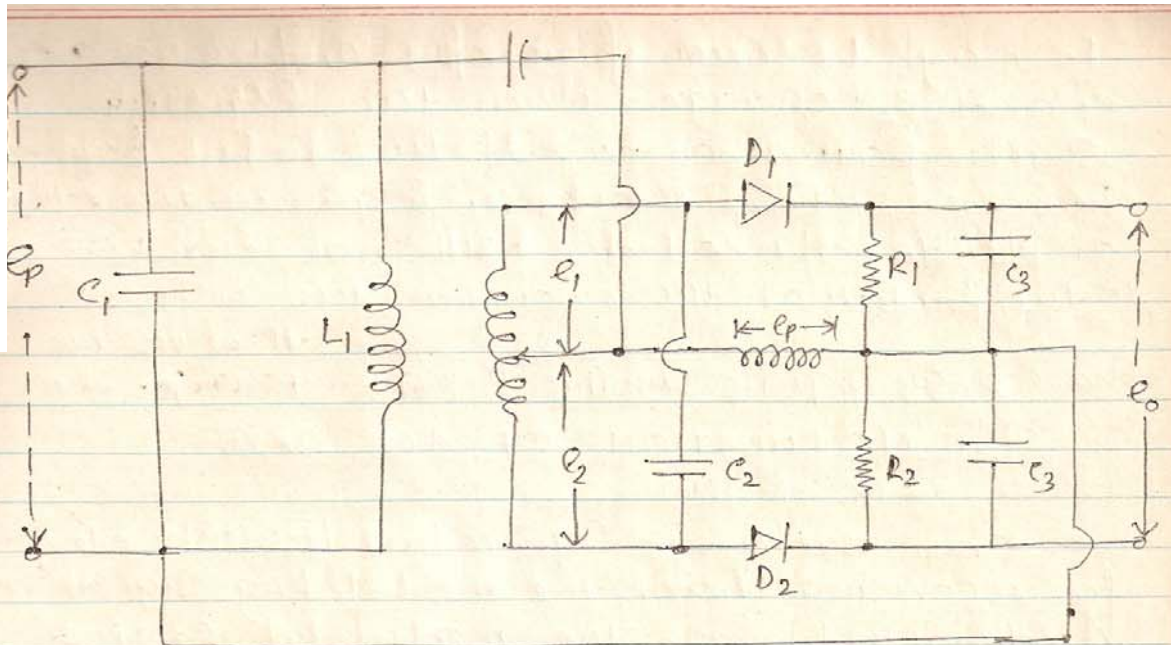
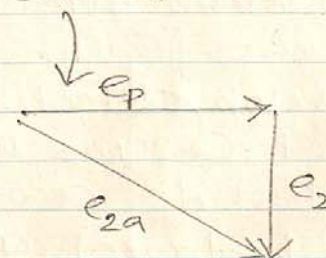
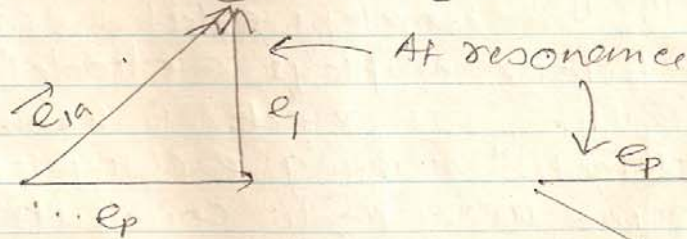


FIG 7: Foster-Seeley discriminator.

e_1 & e_2 by 90° . The voltage e_1 leads the primary voltage e_p by 90° and e_2 lags e_p by 90° when the secondary circuit is tuned to resonance. The voltages e_{1a} and e_{2a} applied to the diode detectors D_1 and D_2 are given by,

$$\vec{e}_{1a} = \vec{e}_1 + \vec{e}_p \quad \text{and}$$

$$\vec{e}_{2a} = \vec{e}_2 + \vec{e}_p$$



2)

FIG 10a

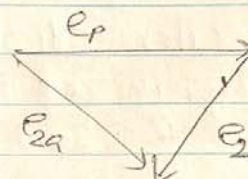
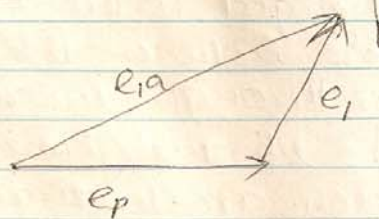


FIG : 10b



increases. This is shown in Fig 10. Similarly for frequency

As may be seen from fig 10, phase angles ϕ_1 and ϕ_2 change when the frequency shifts from resonance. If the input frequency increases above resonance, frequency the relative amplitude of voltages will not change due to the fact that the resonance curve is flat.

However the phase angle is quite sensitive to frequency change. Thus

DEMODULATION OF AM WAVE: (Detection)

We can use any square law transistor device for recovering back the modulating signal (i.e. audio signal) from the modulated carrier. Such a detector is not very efficient.

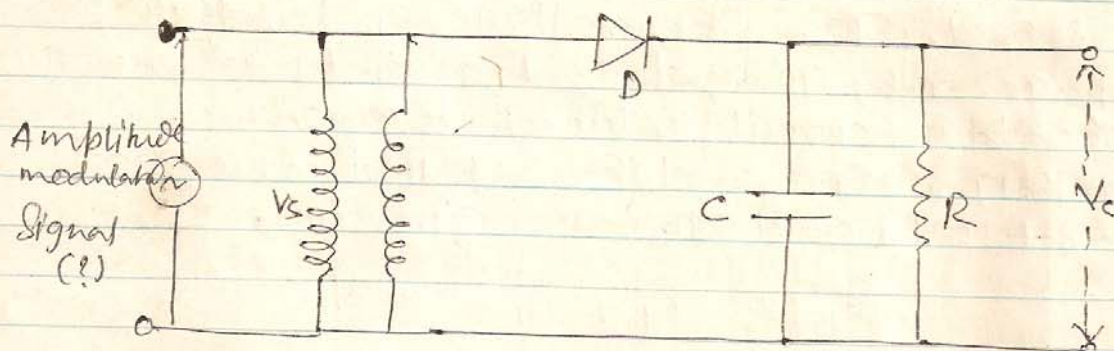


FIG: 11 : Envelop Detector.

A better demodulator employs a diode (D) as shown in fig: 11. Such a ckt. is known as an "envelop detector". During each cycle of the input carrier the diode conducts for a short while and the (capacitor 'C' has low impedance) gets charged to the peak amplitude of the carrier. Between the peaks the capacitor discharges through the resistor (R has high resistance). If the time constant (RC sec) is much greater than the period of the carrier, the discharge of



into amplitude variation the output voltage equals the difference between E_1 and E_2 .

The Capacitor will be very small. The capacitor almost follows the modulation envelope of the input signal. The modulating signal is thus recovered at the detector output.

It can be shown that the time constant should satisfy the following inequality.

$$RC \leq \frac{1}{m_a \omega_m}$$

where

m_a = Index of modulation and

ω_m = maximum frequency of modulation.

from eqⁿ:

$e = E_c (1 + m \sin \omega_m t)$ the modulation envelope is the instantaneous amplitude of an A.M Carrier.

If $RC \leq \frac{1}{\omega_m m_a}$, the distortion in the detector is not excessive.

It is obvious from this equation that the highest frequency that can be detected with reasonable distortion is permissible

$$\omega_{\min} = \frac{1}{RC m_a}$$